

CtrlBooks API Reference Manual

Complete Accounting Synchronization, Data Extraction & AI Execution API

Version: 1.0.1 | **Target Engine:** Tally Prime (HTTP XML Port 9000) | **API Gateway:** Port 8000 | **Date:** September 2026

Document Purpose: This official technical manual provides complete API specifications for Backend Developers and AI Engineers integrating web applications, autonomous AI agents, voice assistants, and enterprise platforms with **CtrlBooks** and local **Tally Prime** instances.

1. Architectural Overview & AI Integration Patterns

Because Tally Prime operates on client desktop machines (behind local network NAT and firewalls on port 9000), external Web applications and Cloud AI assistants cannot establish direct inbound HTTP connections. CtrlBooks resolves this through two production-proven integration patterns:

Pattern A: Cloud 2-Way Command Queue (Recommended for Web AI & Chatbots)

- The Web AI receives natural language (e.g. *"Create sales invoice for Acme Traders for 10 bags Cement"*).
- AI converts intent into a standardized JSON command and queues it in the Cloud database.
- The desktop background agent (`RemoteCommandWorker`) polls for commands every 15 seconds.
- The built-in `ImporterService` posts the XML directly into Tally Prime.
- Execution status, voucher number, and confirmation are returned to the Cloud automatically.

Pattern B: Local REST API Gateway (Port 8000)

For local microservices, desktop agents, or connections tunneled through localhost/VPN, the local API Gateway serves high-performance REST endpoints on `http://localhost:8000`.

2. System Health & Tally Connectivity APIs

Method	Endpoint	Description
GET	/api/system/check	Performs real-time diagnostics: OS compatibility, RAM capacity, Python version, and live connectivity to Tally Prime on port 9000.
GET	/api/system/settings/connection	Returns configured Tally host (default: 127.0.0.1), port (default: 9000), and auto-sync intervals.
POST	/api/system/settings/connection	Updates connection parameters (host IP, port, auto-connect toggle).
GET	/health	Lightweight Gateway health check with sub-millisecond response latency.

Sample Response: GET /api/system/check

```
{
  "status": "GOOD",
  "checks": {
    "os_compatible": true,
    "os_name": "win32",
    "ram_total_gb": 16.0,
    "ram_compatible": true,
    "tally_connected": true
  },
  "version": "1.0.1"
}
```

3. Data Extraction & Query APIs (Read Tally Data)

These endpoints allow AI agents to look up existing company data, customer balances, product stock quantities, and past vouchers before executing

transactions.

Method	Endpoint	Description
GET	/api/v1/website/companies	Lists all synced companies with record count summaries (total ledgers, total vouchers, total stock items).
GET	/api/v1/website/dashboard/summary	Aggregated metrics across all companies for dashboard widgets and high-level KPIs.
GET	/api/v1/website/entities/LEDGER	Queries party and account ledgers (Customers, Vendors, Bank accounts). Supports <code>company_name</code> and <code>search</code> query parameters.
GET	/api/v1/website/entities/STOCK_ITEM	Queries inventory products (Item Name, Closing Stock, Unit, Standard Rate, HSN Code).
GET	/api/v1/website/entities/VOUCHER	Retrieves paginated past transactions (Sales, Purchase, Receipts, Payments) with date and party filters.
GET	/api/v1/website/records/{record_id}	Fetches single raw JSON record details by extracted unique record ID.
POST	/api/v1/website/sync-and-store	Triggers on-demand live extraction from Tally and persists records to the database.

Sample Query: GET /api/v1/website/entities/LEDGER?company_name=Tarun+Enterprise+(25-26)&search=Acme

```
{
  "success": true,
  "data": {
    "total": 1,
```

```
"page": 1,
"page_size": 50,
"items": [
  {
    "id": "rec_67890",
    "entity_type": "LEDGER",
    "name": "Acme Traders",
    "parent": "Sundry Debtors",
    "closing_balance": 24500.0,
    "gstn": "27AAPFU0939F1ZV",
    "mobile": "9876543210",
    "state": "Maharashtra"
  }
]
```

4. AI Transaction Execution Payloads (Write to Tally)

When an AI agent or user requests an action, the AI formats the extracted intention into standard JSON commands. The CtrlBooks engine converts them to Tally XML and executes them truthfully.

A. Sales Invoice (Tax Invoice with Inventory & GST)

Command Type: CREATE_VOUCHER (or Sales)

```
{
  "type": "CREATE_VOUCHER",
```

```
"companyName": "Tarun Enterprise (25-26)",
"payload": {
  "voucher_type": "Sales",
  "party_ledger": "Acme Traders",
  "date": "2026-09-22",
  "narration": "AI Generated Tax Invoice via Voice / Prompt",
  "amount": 11800.0,
  "items": [
    {
      "name": "Cement 50kg",
      "quantity": 10.0,
      "rate": 1000.0,
      "unit": "Bags",
      "amount": 10000.0
    }
  ],
  "ledger_entries": [
    {
      "ledger": "CGST @ 9%",
      "amount": 900.0
    },
    {
      "ledger": "SGST @ 9%",
      "amount": 900.0
    }
  ]
}
```

B. Payment Receipt (Money Received from Customer)

Command Type: CREATE_RECEIPT (or Receipt)

```
{
  "type": "CREATE_RECEIPT",
  "companyName": "Tarun Enterprise (25-26)",
  "payload": {
    "voucher_type": "Receipt",
    "party_ledger": "Acme Traders",
    "date": "2026-09-22",
    "amount": 5000.0,
    "narration": "Payment received via Bank Transfer (Ref #NEFT9823)",
    "ledger_entries": [
      {
        "ledger": "HDFC Bank",
        "amount": 5000.0
      }
    ]
  }
}
```

C. Vendor Payment (Money Paid to Supplier)

Command Type: CREATE_PAYMENT (or Payment)

```
{
  "type": "CREATE_PAYMENT",
  "companyName": "Tarun Enterprise (25-26)",
  "payload": {
    "voucher_type": "Payment",
    "party_ledger": "Tata Steel Ltd",
```

```
{
  "date": "2026-09-22",
  "amount": 25000.0,
  "narration": "Vendor payment against invoice #TS-401",
  "ledger_entries": [
    {
      "ledger": "State Bank of India",
      "amount": 25000.0
    }
  ]
}
```

D. Create New Party Master (Customer / Supplier)

Command Type: `CREATE_LEDGER`

```
{
  "type": "CREATE_LEDGER",
  "companyName": "Tarun Enterprise (25-26)",
  "payload": {
    "name": "Global Infotech Solutions",
    "parent": "Sundry Debtors",
    "state": "Karnataka",
    "gstIn": "29AABCU9603R1ZM",
    "mobile": "9988776655",
    "email": "accounts@globalinfotech.com",
    "address": "100 Feet Ring Road, Bangalore",
    "opening_balance": 0.0
  }
}
```

```
}
```

E. Create New Stock Item (Product Master)

Command Type: CREATE_STOCK_ITEM

```
{
  "type": "CREATE_STOCK_ITEM",
  "companyName": "Tarun Enterprise (25-26)",
  "payload": {
    "name": "Steel Reinforcement Rod 12mm",
    "unit": "Kgs",
    "hsn_code": "72142090",
    "company_name": "Tarun Enterprise (25-26)"
  }
}
```

5. Supported Transaction Voucher Types Matrix

Voucher Type	Command Type Keyword	Accounting Function
Sales	CREATE_VOUCHER / Sales	Customer invoicing with stock allocation & GST tax ledgers.
Purchase	CREATE_PURCHASE / Purchase	Supplier bill entry with inventory addition & input credit.
Receipt	CREATE_RECEIPT / PAYMENT_RECEIVED	Incoming cash, UPI, cheque, or bank receipts.

Payment	CREATE_PAYMENT / PAYMENT_MADE	Outgoing supplier, expense, or payroll payments.
Journal	CREATE_JOURNAL / Journal	General adjustments, depreciation, and year-end entries.
Contra	CREATE_CONTRA / Contra	Cash to bank deposits, cash withdrawals, or inter-bank transfers.
Credit Note	CREATE_CREDIT_NOTE	Sales returns or rate difference allowances to customers.
Debit Note	CREATE_DEBIT_NOTE	Purchase returns or discounts received from vendors.
Sales Order	CREATE_SALES_ORDER	Pending customer purchase orders prior to invoice generation.
Purchase Order	CREATE_PURCHASE_ORDER	Outbound procurement orders sent to vendors.

6. Enterprise Safety & Reliability Safeguards

When developing an AI automation platform, safety and transactional integrity are paramount. CtrlBooks includes the following safeguards out of the box:

Protection Mechanism	How It Protects The Accounting Ledger
Idempotency Guard	Each entry is tracked by a composite hash (<code>company:voucher_type:voucher_number</code>). If a network timeout, web retry, or double prompt occurs, the engine returns the existing voucher confirmation without creating a duplicate in Tally.
Auto-Creation of Masters	If an AI entry references a new customer that does not yet exist in Tally, <code>ImporterService</code> automatically provisions the party ledger under <i>Sundry Debtors</i> with GSTIN details before executing the voucher. The voucher never fails due to missing party errors.
Offline Resilient Queue	If the client turns off their computer or closes Tally Prime, vouchers are safely persisted to the local <code>sync_queue</code> table.

	When Tally is reopened, the background agent automatically drains and posts all pending entries.
Penny Balancing	Automatically compensates for floating-point rounding discrepancies between item tax subtotals and invoice totals to ensure zero ledger rejections.

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